Project - Low Level Design
on
Mutli – Agent Insurance System

Course Name: AAI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1.	YASH SHRIVASTAVA	EN22CS3011111
2.	VISHWAS BHARGAV	EN22CS3011095
3.	YASHI HARNE	EN22CS3011116
4.	DIXIKA VERMA	EN22ME304027
5.	ANVESHA WANKHADE	EN22ME304014

Group Name: Group 07D10

Project Number: AAI - 20

Industry Mentor Name:

University Mentor Name: Prof. Maya Baniya

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Table of Contents

1. Introduction.
   1.1. Scope of the document.
   1.2. Intended Audience
   1.3. System overview.

2. System Design.
   2.1. Application Design
   2.2. Process Flow.
   2.3. Information Flow.
   2.4. Components Design
   2.5. Key Design Considerations
   2.6. API Catalogue.

3. References

1. Introduction

The Multi-Agent Insurance System is an AI-powered system that uses multiple intelligent agents working collaboratively to perform complex insurance-related tasks. Instead of relying on a single AI model, this system divides responsibilities among specialized agents that communicate and coordinate with each other.

The system includes:

- Researcher Agent – Responsible for collecting insurance policy information such as coverage, eligibility, and benefits.
- Writer Agent – Responsible for organizing, summarizing, and presenting the gathered information in a structured format.

Using frameworks such as LangChain or CrewAI, the agents communicate with each other, share context, and hand off tasks to ensure high-quality output. The system demonstrates the concept of Agentic AI, where autonomous agents collaborate to solve complex problems.

This approach improves accuracy, modularity, and scalability compared to traditional AI systems.

1.1. Scope of the Document

This document provides a detailed description of the architecture, design, and workflow of the Multi-Agent Insurance System.

It covers:

- System architecture and components
- Agent communication and collaboration
- Process flow and information flow
- Design considerations and system decisions
- API structure used for agent interaction

The document serves as a guide for developers, researchers, and stakeholders to understand how the system operates and how it can be implemented.

1.2. Intended Audience

This document is intended for:

- Insurance professionals seeking clear, visual representations of complex concepts.
- AI developers and data scientists implementing generative AI workflows.
- Educators and trainers designing instructional content.
- Project managers and stakeholders overseeing AI-based innovation initiatives.

It ensures that all readers understand the system's capabilities, relevance, and potential applications.

1.3. System Overview

The Multi-Agent Insurance System consists of multiple AI agents that collaborate to process insurance queries and generate structured responses.

The system workflow is as follows:

1. A user submits an insurance-related query.
2. The Researcher Agent gathers relevant insurance information from knowledge sources or databases.
3. The gathered information is passed to the Writer Agent.
4. The Writer Agent analyzes, summarizes, and formats the information.
5. The system returns a structured response to the user.

Key Features

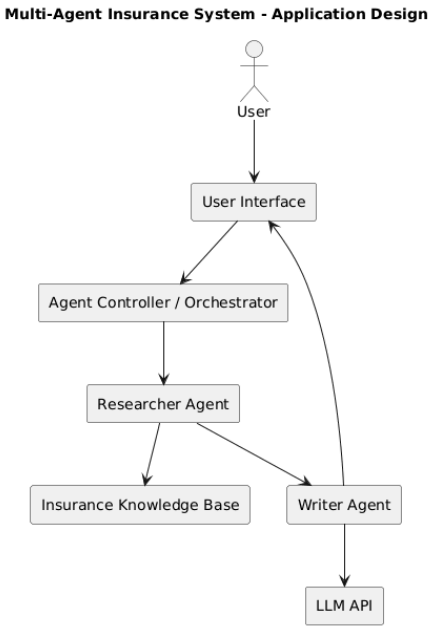
- Multi-agent collaboration
- Autonomous task delegation
- Context sharing between agents
- Structured content generation
- Scalable architecture

This architecture allows each agent to specialize in a specific task, improving

overall system performance.

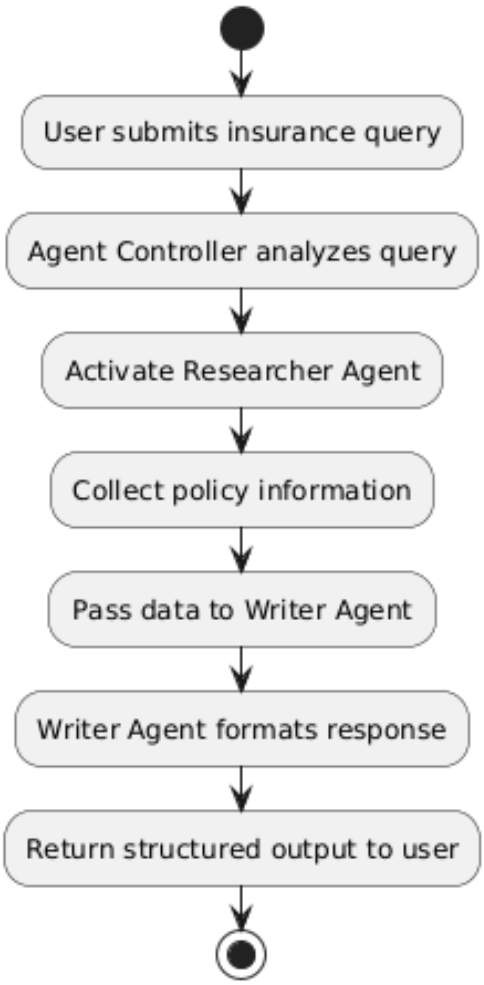
2. System Design.

2.1Application Design



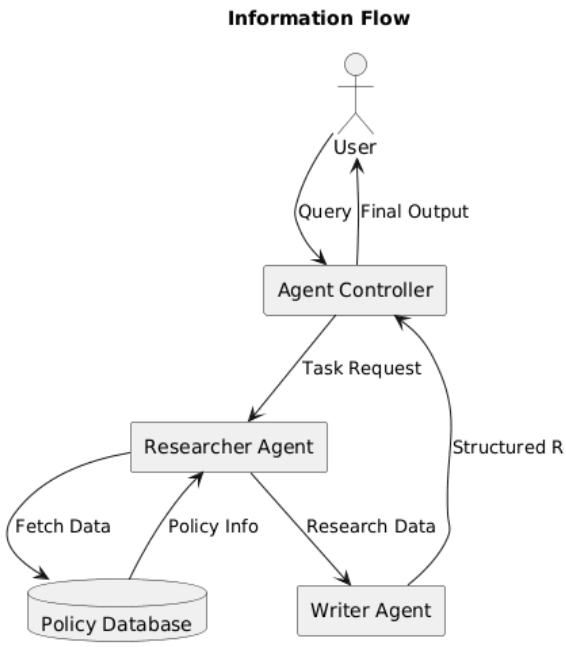
- User submits a query through the User Interface.
- The Agent Controller manages workflow between agents.
- Researcher Agent collects policy data from knowledge sources.
- Data is sent to the Writer Agent.
- Writer Agent uses LLM API to generate structured responses.
- The final response is returned to the user.

2.2Process Flow



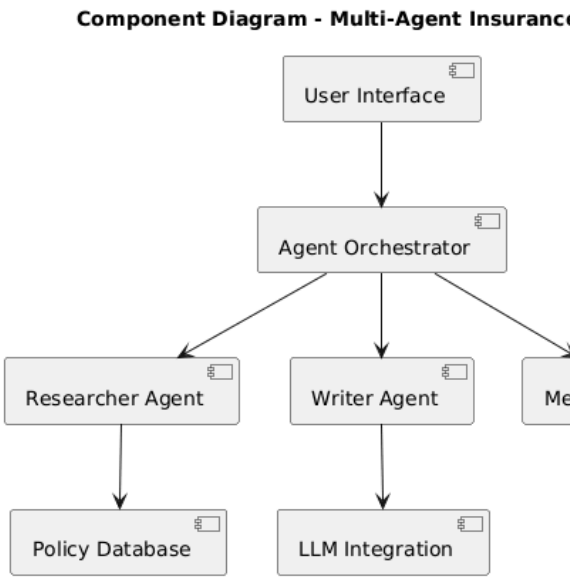
- The workflow starts with a user query.
- The system analyzes the request.
- The Researcher Agent retrieves relevant information.
- Information is transferred to the Writer Agent.
- The writer agent organizes and formats the data.
- The system sends the final answer to the user.

2.3 Information Flow.



- User sends a query to the system.
- The Agent Controller assigns tasks to the Researcher Agent.
- The researcher retrieves policy data from the database.
- Collected data is passed to the Writer Agent.
- Writer Agent creates structured content.
- The system returns the final response to the user.

2.4 Components Design



- User Interface interacts with users.
- Agent Orchestrator coordinates system activities.
- Researcher Agent collects insurance information.
- Writer Agent generates readable outputs.
- LLM Integration provides language intelligence.
- Memory System stores conversation context.
- Policy Database contains insurance data.

2.5 Key Design Considerations

Several design factors were considered when developing the system.

1. Modularity
Each agent is designed independently, allowing easy upgrades or modifications.

2. Scalability
The architecture allows additional agents such as:
- Risk Analysis Agent
- Policy Recommendation Agent
- Fraud Detection Agent

3. Fault Tolerance

If one agent fails, the system can retry or reassign tasks.

4. Context Management
Maintaining context ensures agents understand previous steps and avoid redundant processing.

5. Security
Sensitive insurance information must be handled securely with proper data protection mechanisms.

6. Performance Optimization
Efficient agent communication reduces processing time and improves response quality.

2.6 API Catalogue:

The API catalogue defines the interfaces used for agent communication.

1. Query Submission API

- **Endpoint**

POST /query

- **Description**

Accepts user queries related to insurance policies.

- **Input**

{

"query": "What does health insurance policy cover?"

}

- Output

{

"response": "Health insurance policies generally cover hospitalization, medical treatments, and prescribed medications."

}

2. Research Agent API

- Endpoint

POST /research

- Description

Retrieves insurance policy information.

- Input

```
{
  "topic": "health insurance coverage"
}
```

- Output

```
{

  "research_data": "Health insurance covers hospitalization, surgery, and medical expenses."

}
```

3. Writer Agent API

- Endpoint

POST /write

- Description

Formats research data into structured content.

- Input

```
{
  "data": "health insurance covers hospitalization..."
}
```

- Output

```
{

  "formatted_content": "Health insurance policies provide financial protection against medical expenses..."

}
```

3.References

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